Toronto Bicycle Theft Data

Spatiotemporal Distribution and Spatial Interpolation Method Analysis Report

Data period: 2014–2023 | Sample size: 31,833 records | Source: Toronto Police Service open data

1. Dataset Overview

This dataset records 31,833 bicycle theft incidents in the City of Toronto between 2014 and 2023. Fields include the incident date, quarter, day of week, neighborhood name, value of the stolen bicycle, premises type, and geographic coordinates. There are no missing values, and the time span fully covers 10 calendar years.

Key Summary Statistics

- Time range: 2014-01-01 to 2023-12-31, spanning 10 years
- Neighborhoods covered: 140 official City of Toronto neighborhoods
- Premises types: Residential Structures (47.5%), Commercial Areas (25.9%), Open/Public Spaces (20.2%), and 4 other categories
- Stolen bike value: median CAD 650, mean CAD 994, maximum CAD 120,000 (including outliers)

2. Temporal Distribution (by Quarter)

When all thefts are grouped by calendar quarter, a strong seasonal pattern emerges that closely tracks Toronto's cycling season.

Quarter	Thefts	Share	Characteristics
Q1 (Jan–Mar)	2,631	8.3%	Lowest volume of the year; little cycling in winter
Q2 (Apr–Jun)	9,512	29.9%	Warming weather drives a rapid rise in cycling and thefts
Q3 (Jul–Sep)	13,728	43.1%	Annual peak quarter, accounting for over four-tenths
Q4 (Oct–Dec)	5,962	18.7%	Noticeable decline as winter approaches

Table 1: Theft counts by quarter, 2014–2023

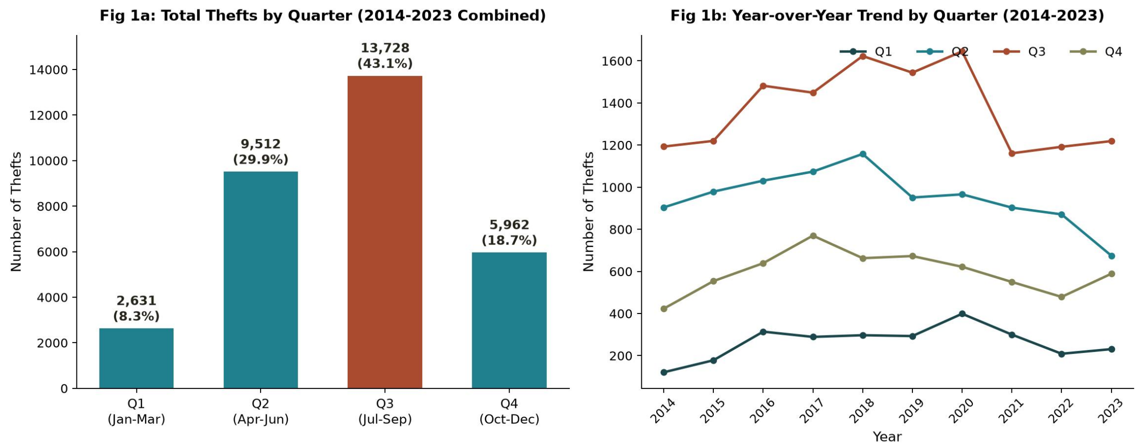

Figure 1: Total thefts by quarter (left) and year-over-year trend by quarter, 2014–2023 (right)

Key Findings

- Strong concentration in warm months: Q3 (Jul–Sep) recorded 13,728 thefts, or 43.1% of the annual total — the clear peak. Q2 (Apr–Jun) accounts for 29.9%; together Q2 and Q3 make up roughly 73% of all thefts.
- Lowest volume in winter: Q1 (Jan–Mar) accounts for only 8.3% of thefts, reflecting the sharp drop in cycling and greater use of indoor bike storage during winter.
- Seasonal pattern is stable year over year: the year-by-year breakdown from 2014 to 2023 shows the Q3 peak persisting every single year, indicating a structural seasonal pattern rather than a one-off fluctuation.
- Overall volume rose then declined: total thefts increased steadily from 2014 to 2018 (2,641 → 3,741), then trended down with fluctuations after 2019, settling around 2,700–2,900 per year in 2021–2023. This decline may relate to post-pandemic cycling habits, policing resource allocation, or improved lock technology, and would benefit from further validation against external context.
- Stolen bike value shows a counter-seasonal pattern: Q1 has the fewest thefts but the highest median value (CAD 800), while Q3 has the most thefts but the lowest median value (CAD 600) — suggesting winter thefts more often involve higher-value commuter/specialty bikes.

3. Spatial Distribution Characteristics

At the neighborhood level, thefts show a pronounced core-periphery structure: a small number of downtown neighborhoods are highly concentrated, while the remaining 120 neighborhoods share less than four-tenths of all cases.

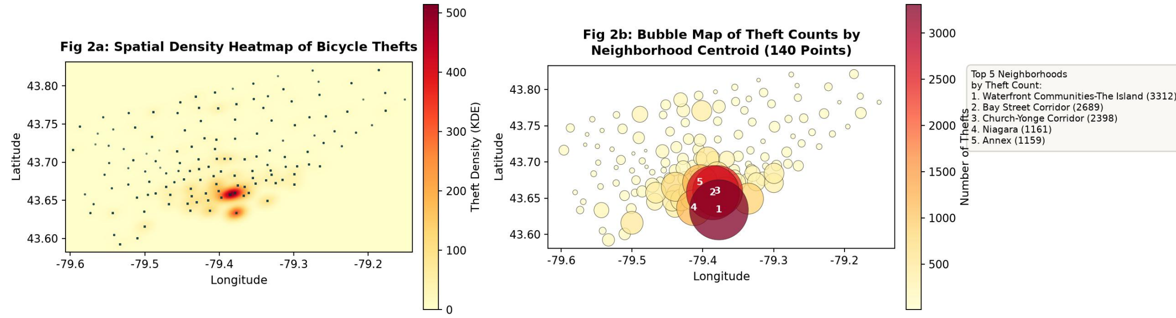

Figure 2: Spatial kernel density heatmap of thefts (left) and bubble map of theft counts by 140 neighborhood centroids (right)

Key Findings

- Highly clustered spatial distribution: the top 10 neighborhoods (7% of all 140 neighborhoods) together account for 48.3% of all thefts; the top 20 account for 62.9%; the remaining 120 neighborhoods share only about 37%.
- Three major downtown hotspots: Waterfront Communities-The Island (3,312), Bay Street Corridor (2,689), and Church-Yonge Corridor (2,398) form the densest cluster, together accounting for 25.9% of citywide thefts.
- Single-core radial decay pattern: the kernel density map shows hotspots concentrated in Toronto's downtown waterfront and midtown core, with density decaying rapidly toward the suburbs — consistent with typical urban-core crime clustering.
- Premises type correlates with location: dense downtown neighborhoods (e.g., Bay Street, Church-Yonge) show a higher share of commercial/public-space thefts, while peripheral neighborhoods are dominated by residential-structure thefts, reflecting differing land-use functions.

4. Spatial Distribution of Stolen Bike Value

Beyond theft frequency, the value of stolen bicycles also shows a spatial pattern that differs markedly from theft density — warranting separate analysis.

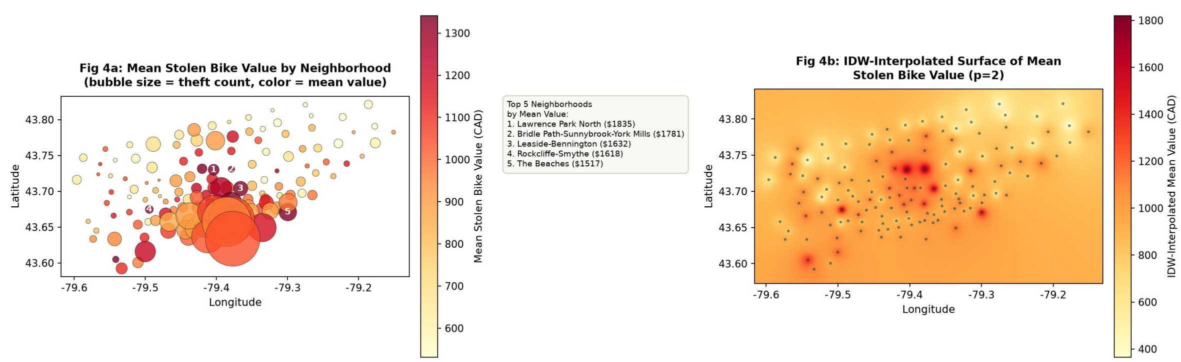

Figure 3: Bubble map of mean stolen bike value by neighborhood (left, bubble size = theft count, color = mean value) and IDW-interpolated surface (right, p=2)

Key Findings

- High-value neighborhoods cluster in affluent mid-north areas, not in the busiest downtown core: the five neighborhoods with the highest mean values are Lawrence Park North (CAD 1,835), Bridle Path-Sunnybrook-York Mills (CAD 1,781), Leaside-Bennington (CAD 1,632), Rockcliffe-Smythe (CAD 1,618), and The Beaches (CAD 1,517) — all traditional higher-income residential areas north or east of midtown.
- Theft count and mean value are barely correlated: the correlation coefficient is only 0.14, meaning the neighborhoods with the most thefts (e.g., Waterfront Communities, Bay Street Corridor) are not the ones with the highest-value losses — their high-density, high-turnover commercial/public spaces tend to see mostly lower- to mid-value commuter bikes stolen.
- Low-value neighborhoods concentrate in the northeast and eastern periphery: areas such as Humbermede (CAD 355), Highland Creek (CAD 380), and Milliken (CAD 396) have fewer thefts but also consistently lower stolen bike values.
- The IDW-interpolated surface (Fig. 3, right) clearly shows a high-value band extending from the low-value southwest toward the mid-north (Lawrence Park, Bridle Path area), spatially offset from the theft-density hotspot in Fig. 2 (concentrated in the downtown waterfront). This indicates that "high-frequency areas" and "high-value-loss areas" are two distinct dimensions for policy attention, and resource allocation should separately consider incident frequency and financial loss severity.

5. Key Data Characteristic: Coordinates Represent Neighborhood Centroids, Not Exact Addresses

Before selecting a spatial interpolation method, it is essential to flag a key structural feature of this dataset: the latitude/longitude values do not represent the exact address of each incident — they are neighborhood centroid coordinates.

- All 31,833 records map to only 140 unique coordinate pairs, which correspond exactly one-to-one with the 140 neighborhood names — meaning every theft within a given neighborhood shares the identical coordinate pair.
- This means the theft data is, spatially, fundamentally areal aggregate count data (choropleth in nature) rather than independent random samples from a continuous spatial field.
- This characteristic directly determines the interpolation method selection logic discussed in Section 6.

6. Spatial Interpolation Method Selection: IDW or Kriging?

Recommendation: prioritize IDW (Inverse Distance Weighting); ordinary Kriging is not recommended as a direct choice.

Rationale

- Too few effective sample points, and highly repeated: the true number of spatial "observations" is only 140 (neighborhood centroids), not 31,833 independent locations. Kriging relies on a variogram to characterize spatial autocorrelation, which requires a sufficient number of fairly randomly distributed independent sample points to fit reliably. 140 regularly arranged, repeated points make variogram estimation unstable and prone to directional artifacts.
- The data is fundamentally areal aggregate data, not a point process: since the coordinates are neighborhood representative points, theft counts should be understood as zonal counts rather than samples from a continuous spatial field. Point interpolation therefore has limited geographic meaning here, and IDW can serve as a more pragmatic visualization approach.
- IDW's assumptions are much lighter: it only relies on a distance-decay assumption, with no requirement for a spatial autocorrelation structure. It has few parameters (power exponent p, search radius), is computationally simple, and is more robust when point locations are regular and sample size is limited.
- Kriging's advantages are hard to realize here: its core strengths — unbiased optimal estimation plus interpolation uncertainty (variance maps) — depend on reliably fitting a variogram. With 140 fixed, repeated points, there isn't enough pairwise distance-gradient information across varying distances, so this advantage cannot be meaningfully realized on this dataset.

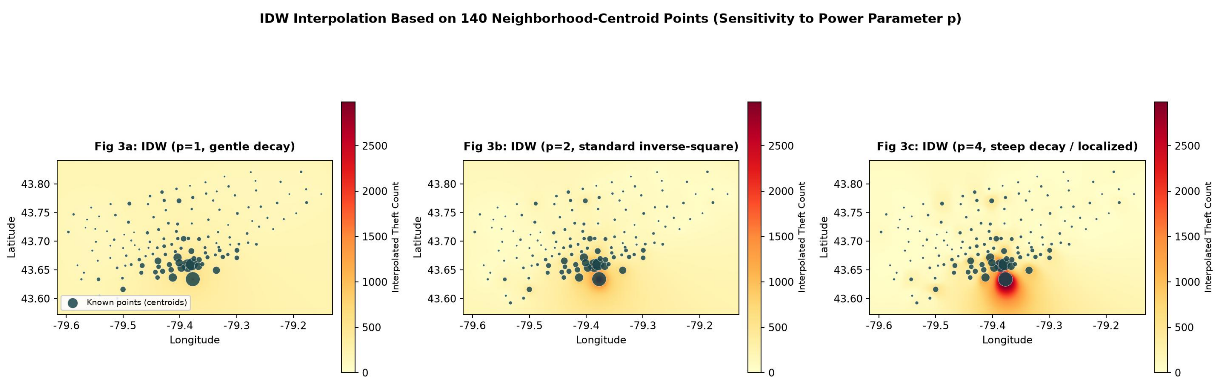

Figure 4: IDW interpolation example based on 140 known neighborhood-centroid points (power parameter p = 1, 2, 4 compared). As p increases, the interpolated surface concentrates more sharply around individual points (e.g., an isolated "bullseye" appears near Church-Yonge/Bay Street at p=4), highlighting IDW's sensitivity to parameter choice and, indirectly, the effect of repeated centroid points on interpolation stability.

7. IDW vs. Kriging: Pros and Cons

Dimension	IDW (Inverse Distance Weighting)	Kriging
Theoretical basis	Deterministic method: weight increases as distance decreases; no statistical assumptions	Geostatistical method: based on regionalized variable theory and spatial autocorrelation (variogram)
Data requirements	Low requirements on sample size and spatial regularity	Requires sufficient, fairly random/uniform sample points to reliably fit the variogram
Computational	Simple; only requires setting the	Requires fitting a variogram model

Dimension	IDW (Inverse Distance Weighting)	Kriging
complexity	power parameter p and search radius; fast	(spherical/exponential/Gaussian, etc.); complex parameter tuning
Accuracy / optimality	Not optimal; ignores directionality and clustering in spatial structure	Best Linear Unbiased Estimator (BLUE) in theory; can capture anisotropy
Uncertainty quantification	Cannot provide prediction variance / confidence intervals	Can output kriging variance, quantifying prediction uncertainty at each point
Robustness to duplicate/clustered points	Fairly robust; duplicate points simply add weight at that location	Sensitive; duplicate points distort the variogram estimate, risking overfitting or ill-conditioned matrices
Sensitivity to parameter choice	Sensitive to the power parameter p, but its effect is intuitive and easy to tune	Highly sensitive to variogram model, sill, and range choices; high tuning threshold
Suitability for this dataset	√ Better suited: 140 regular, repeated points; no variogram assumption required; robust	△ Not ideal: variogram is hard to fit reliably; interpolated surface may be distorted

Table 2: Comprehensive comparison of IDW and Kriging

Bottom line: Kriging delivers superior accuracy and uncertainty characterization when samples are abundant and spatially random, but it is poorly suited to this dataset's sparse, regularized point structure where centroids represent large numbers of repeated events. IDW's simplicity and lighter assumptions make it the more reliable choice here. For a more rigorous approach, consider reprocessing the data as areal data and applying spatial autocorrelation statistics such as Moran's I or neighborhood-level density classification maps, rather than forcing point interpolation.

8. Conclusions and Recommendations

- Temporal dimension: thefts show strong seasonality, with Q3 (summer) as the priority period for prevention — recommend increased patrols, public awareness campaigns, and bike registration drives from April through September.
- Spatial dimension (frequency): thefts are heavily concentrated in a handful of downtown neighborhoods (Waterfront Communities, Bay Street Corridor, Church-Yonge Corridor, etc.) — recommend prioritizing surveillance infrastructure, secure parking upgrades, or dedicated patrols in these hotspots.
- Spatial dimension (value): high-value-loss areas (Lawrence Park North, Bridle Path, and other mid-north residential neighborhoods) are spatially offset from high-frequency theft areas — recommend targeted measures such as bike registration and insurance outreach specifically for these high-value-loss areas.

- Methodology: given that coordinates represent neighborhood centroids rather than true addresses, IDW is the more appropriate spatial interpolation method over Kriging; if more precise incident-level addresses become available in the future, Kriging or kernel density estimation could be re-evaluated as more rigorous geostatistical alternatives.
- Further directions: incorporating covariates such as population density, cycling-infrastructure network density, and transit-stop distribution could support Regression Kriging or Geographically Weighted Regression (GWR) models for improved spatial prediction accuracy.